A REPORT

ON

ARTIFICIAL INTELLIGENCE

Submitted by

Ms. Keerthi M S – 20241CSG0010

Under the guidance of,

Mr .Kavin Mahendhiran
Mr. Asad Mohammed Khan

in partial fulfillment for the award of the

degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING (COMPUTER ENGINEERING)

AT

GAIN MORE KNOWLEDGE
REACH GREATER HEIGHTS

PRESIDENCY UNIVERSITY

BENGALURU

AUGUST 2026

PRESIDENCY UNIVERSITY

PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

This is to certify that the report of **CSE7000 – Internship** on **"ARTIFICIAL INTELLIGENCE "** being submitted by "**KEERTHI M S**" bearing roll number "**20241CSG0010**" in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in **Computer Science and Engineering (Computer Science and Technology)** is a bonafide work carried out under our supervision.

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DECLARATION

I hereby declare that the work, which is being presented in the report entitled "**ARTIFICIAL IN**TELLIGENCE**"** in partial fulfillment for the award of Degree of **Bachelor of Technology** in **Computer Science and Engineering (Computer Science and Technology),** is a record of my own investigations carried under the guidance of**. Mr. Kavin Mahendhiran, **Assistant Professor and**. Mr. Asad Mohammed Khan ,** Assistant Professor , Presidency School of Computer Science and Engineering, Presidency University, Bengaluru.

I have not submitted the matter presented in this report anywhere for the award of any other Degree.

Name, Roll No and Signature of the Student

INTERNSHIP COMPLETION CERTIFICATE


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CERTIFICATE
OF COMPLETION

THIS CERTIFICATE IS PRESENTED TO

Keerthi MS

In recognition of his/her efforts and achievements in completing
1 Month internship in Artificial Intelligence.
Conducted From 1st June 2026 to 30th June 2026

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MSME

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First of all, I indebted to the **GOD ALMIGHTY** for giving me an opportunity to excel in our efforts to complete this internship on time.

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Name of the Student (Roll No)

ABSTRACT

The internship in Artificial Intelligence (AI) provided an opportunity to gain practical knowledge and hands-on experience in one of the rapidly developing areas of computer science. Artificial Intelligence is a field of technology that focuses on developing computer systems capable of performing tasks that normally require human intelligence, such as learning, reasoning, decision-making, pattern recognition, prediction, and problem solving. The internship helped in understanding the fundamental concepts of Artificial Intelligence and their applications in real-world situations.

During the internship, the learning process focused on understanding the basic concepts and workflow involved in developing AI-based solutions. The internship introduced important topics such as Artificial Intelligence, Machine Learning, data preprocessing, algorithms, model training, testing, prediction, and evaluation. The theoretical concepts were supported by practical exercises and implementation-based activities, which helped in developing a better understanding of how AI techniques can be applied to solve computational problems.

One of the important aspects of the internship was understanding the relationship between data and Artificial Intelligence. Data is an essential component of AI systems because machine learning models learn patterns and relationships from available data. The internship provided an understanding of how data can be collected, prepared, processed, and used for developing intelligent systems. Data preprocessing activities such as handling data, organizing datasets, identifying relevant features, and preparing data for model development were studied as part of the

learning process.

The internship also provided exposure to the basic workflow of a Machine Learning project. This includes identifying a problem, collecting or selecting suitable data, preprocessing the data, selecting an appropriate algorithm, training the model, testing its performance, and evaluating the results. Through these activities, the importance of selecting suitable algorithms and preparing quality data was understood. The internship also helped in understanding that the performance of an AI system depends on several factors, including the quality of data, choice of algorithm, training process, and evaluation method.

Practical implementation was an important part of the internship. Programming and development activities helped in converting theoretical concepts into practical solutions. Working with AI-related tools and programming environments improved technical skills and provided an understanding of how software development concepts are integrated with Artificial Intelligence. The practical exercises also helped in improving logical thinking, problem-solving ability, debugging skills, and programming knowledge.

The internship further provided an introduction to different applications of Artificial Intelligence. AI can be applied in areas such as healthcare, education, finance, transportation, recommendation systems, automation, cybersecurity, and many other domains. Understanding these applications helped in recognizing the importance of AI in modern software systems and its potential to solve complex real-world problems.

Another important outcome of the internship was the development of

professional and technical skills. The internship encouraged systematic problem solving, experimentation, documentation, and analysis of results. It also helped in understanding the importance of learning continuously in the field of Artificial Intelligence because AI technologies and methods are constantly evolving.

Overall, the Artificial Intelligence internship provided a valuable combination of theoretical understanding and practical exposure. It strengthened the understanding of AI and Machine Learning concepts and provided a foundation for further learning and development in the field. The knowledge gained during the internship can be applied to future academic projects, mini-projects, and software development activities. The internship also helped in understanding the skills required to work on AI-based applications and contributed to the development of confidence in approaching technical problems using computational and intelligent methods.

INTRODUCTION

Artificial Intelligence (AI) is a branch of computer science that focuses on developing systems capable of performing tasks that normally require human intelligence. These tasks include learning from data, recognizing speech, understanding natural language, solving problems, making decisions, and recognizing images. AI has become one of the most influential technologies in the modern world because of its ability to automate complex processes and improve efficiency.

The rapid growth of digital data and computing power has accelerated AI development over the past decade. Organizations across various industries use AI to analyze large volumes of data, predict outcomes, automate repetitive tasks, and improve customer experiences. AI-powered applications such as virtual assistants, recommendation systems, autonomous vehicles, medical diagnosis systems, and fraud detection systems are now part of everyday life.

Machine Learning (ML), a subset of AI, enables computers to learn from data without being explicitly programmed. ML algorithms identify patterns in data and use these patterns to make predictions or decisions. Deep Learning, an advanced branch of Machine Learning, utilizes artificial neural networks with multiple hidden layers to solve complex problems involving image recognition, speech processing, and natural language understanding.

The objective of this internship was to gain practical exposure to Artificial Intelligence technologies through theoretical learning and hands-on implementation. Various AI algorithms were implemented using Python programming language and industry-standard libraries such as Scikit-learn, TensorFlow, Pandas, NumPy, and Matplotlib.

The internship provided valuable experience in data collection, preprocessing, feature engineering, model development, training, testing, evaluation, and deployment. It also enhanced programming skills, logical thinking, and understanding of real-world AI applications.

Artificial Intelligence continues to evolve rapidly and is expected to play a significant role in the future of healthcare, education, finance, robotics, smart cities, cybersecurity, agriculture, and industrial automation. Learning AI technologies during this internship has built a strong foundation for advanced research and professional development in the field of intelligent computing.

ABOUT THE COMPANY / ORGANIZATION

Company Overview

Alfido Tech Pvt. Ltd. is a technology-driven organization specializing in Artificial Intelligence (AI), Machine Learning (ML), Data Science, Cloud Computing, and Software Development. The company aims to provide innovative AI-powered solutions that help businesses automate processes, improve decision-making, and enhance productivity.
Established with the vision of promoting advanced digital technologies,

the organization delivers industry-oriented training, research, and software solutions. It focuses on bridging the gap between academic learning and industrial requirements by offering internship programs, live projects, and technical workshops.

The company has a team of experienced software engineers, AI researchers, and data scientists who work on developing intelligent systems for various industries, including healthcare, finance, agriculture, education, retail, and manufacturing. During the internship, students receive practical exposure to modern AI technologies through hands-on projects and real-world datasets.

The organization encourages innovation, teamwork, and continuous learning by allowing interns to participate in project discussions, coding sessions, and model development activities. It also emphasizes ethical AI development, data security, and responsible use of technology.

Vision:

To become a leading organization in Artificial Intelligence research, innovation, and technology solutions that contribute to sustainable digital transformation.

Mission:

- **Deliver high-quality AI solutions for industry applications.**
- **Promote research and innovation in Artificial Intelligence.**
- **Train students with practical and industry-relevant skills.**
- **Encourage ethical and responsible AI development.**
- **Bridge the gap between academics and industry through experiential learning.**

WORKING DOMAIN / TECHNOLOGY

The internship was conducted in the domain of Artificial Intelligence, with a primary focus on Machine Learning, Deep Learning, Data Science, and Intelligent Systems.

Artificial Intelligence is the science of designing systems that can imitate human intelligence by learning from data, recognizing patterns, making predictions, and solving complex problems. Modern AI systems are capable of continuously improving their performance through learning algorithms.

Technologies Covered During the Internship

1. Python Programming

Python was used as the primary programming language because of its simple syntax and extensive support for AI libraries. Students learned programming concepts including variables, loops, functions, object-oriented programming, and file handling before implementing AI algorithms.

2. Machine Learning

Machine Learning enables computers to learn from historical data without explicit programming. Various supervised and unsupervised learning algorithms were studied.

Examples include:

- **Linear Regression**
- **Logistic Regression**
- **Decision Tree**
- **Random Forest**
- **Support Vector Machine**
- **K-Nearest Neighbor**
- **Naïve Bayes**
- **K-Means Clustering**

3. Deep Learning

Deep Learning is a subset of Machine Learning that uses Artificial Neural Networks with multiple hidden layers for solving complex problems.

Topics covered include:

- **Artificial Neural Networks (ANN)**
- **Convolutional Neural Networks (CNN)**

- **Recurrent Neural Networks (RNN)**
- **Model Optimization**
- **Image Classification**

4. Data Science

Data Science involves collecting, cleaning, analyzing, and visualizing data before developing AI models.

Activities included:

- **Data Collection**
- **Data Cleaning**
- **Handling Missing Values**
- **Feature Engineering**
- **Data Visualization**
- **Statistical Analysis**

5. Natural Language Processing (NLP)

Natural Language Processing enables computers to understand and process human language.

Applications include:

- **Sentiment Analysis**
- **Chatbots**
- **Language Translation**
- **Text Classification**
- **Spam Detection**
- **Speech Recognition**

6. Computer Vision

Computer Vision enables machines to interpret and analyze digital images.

Applications include:

- **Face Recognition**
- **Image Classification**
- **Object Detection**
- **Medical Image Analysis**
- **Traffic Monitoring**

INTERNSHIP WORK PROGRESS

The internship was conducted over several weeks, with each week focusing on different Artificial Intelligence concepts and practical activities.

Week 1: Introduction to Artificial Intelligence

- **Understanding AI fundamentals**
- **History and evolution of AI**
- **Overview of Machine Learning and Deep Learning**
- **Installing Python and development tools**

Week 2: Python Programming

- **Variables and data types**
- **Control statements**
- **Functions**
- **Object-oriented programming**
- **File handling**

Week 3: Data Analysis and Preprocessing

- **Loading datasets using Pandas**
- **Handling missing values**
- **Data visualization with Matplotlib**
- **Feature engineering**
- **Data normalization**

Week 4: Machine Learning

- **Regression algorithms**
- **Classification algorithms**
- **Clustering techniques**
- **Model training**
- **Model testing**

Week 5: Deep Learning

- **Neural Networks**
- **TensorFlow and Keras**
- **CNN architecture**
- **Image classification**
- **Hyperparameter tuning**

Week 6: NLP and Computer Vision

- **Text preprocessing**
- **Sentiment analysis**
- **Image processing with OpenCV**
- **Object detection basics**
- **Model evaluation and documentation**

AI PROJECT IMPLEMENTATION

Project Overview:

As part of the Artificial Intelligence internship, a practical project titled "Student Performance Prediction Using Machine Learning" was developed. The objective of the project was to analyze student-related data and predict academic performance based on several influencing factors such as attendance, study hours, previous examination scores, assignment completion, and participation in classroom activities.

The project demonstrated the complete lifecycle of an AI application, including data collection, preprocessing, model training, testing, evaluation, and prediction. It provided practical exposure to applying machine learning algorithms to solve real-world educational problems.

Project Objectives:

The primary objectives of the project were:

- To collect and preprocess educational datasets.
- To understand the relationship between different academic factors and student performance.
- To build predictive machine learning models.
- To compare the performance of different algorithms.
- To evaluate model accuracy using standard evaluation metrics.
- To gain practical experience in AI model development.

Project Development Process

The project followed these development stages:

1. Problem Identification

The problem was defined as predicting whether a student is likely to perform well based on historical academic data.

2. Data Collection

A structured dataset containing student records was collected. The dataset included attributes such as:

- **Student ID**
- **Attendance Percentage**
- **Study Hours**
- **Assignment Scores**
- **Previous Semester Marks**
- **Internal Assessment Scores**

3. Data Preprocessing

The collected data was cleaned by:

- **Removing duplicate records**
- **Filling missing values**
- **Encoding categorical variables**
- **Normalizing numerical data**
- **Splitting data into training and testing datasets**

4. Model Training

Several machine learning algorithms were trained, including:

- **Decision Tree**
- **Random Forest**
- **Logistic Regression**

- **Support Vector Machine (SVM)**
- **K-Nearest Neighbors (KNN)**

5. Model Testing

The trained models were tested using unseen data to evaluate prediction performance.

6. Performance Evaluation

The models were compared using:

- **Accuracy**
- **Precision**
- **Recall**
- **F1-Score**
- **Confusion Matrix**

INTERNSHIP EXPERIENCE

The Artificial Intelligence internship was a valuable learning experience that enhanced both technical and professional skills. It provided practical exposure to modern AI technologies, programming tools, and machine learning frameworks. Working with datasets, building predictive models, and evaluating AI systems helped improve analytical thinking and problem-solving abilities.

The internship also emphasized teamwork, technical documentation, effective communication, and time management. Guidance from mentors and regular practical assignments encouraged continuous learning and increased confidence in implementing AI solutions independently.

Overall, the internship successfully bridged the gap between academic knowledge and real-world applications, preparing me for future opportunities in Artificial Intelligence, Machine Learning, Data Science, and Software Development.

RESULTS

Overview

The Artificial Intelligence internship successfully achieved its intended learning objectives by providing both theoretical knowledge and practical experience in AI technologies. Through structured training, programming exercises, and project implementation, a comprehensive understanding of AI concepts and their real-world applications was developed.

The internship emphasized the complete lifecycle of AI model development, including data collection, preprocessing, model training, testing, evaluation, and performance optimization. Practical implementation using Python and modern AI frameworks significantly improved technical competence and confidence.

Technical Outcomes:
The following technical outcomes were achieved during the internship:

1. Python Programming Skills
A strong understanding of Python programming was developed, including:

- **Variables and data types**
- **Control structures**
- **Functions and modules**
- **Object-Oriented Programming**
- **File handling**
- **Exception handling**

2. Machine Learning Implementation

Several machine learning algorithms were implemented successfully using Scikit-learn.

Algorithms studied included:

- **Linear Regression**
- **Logistic Regression**
- **Decision Tree**
- **Random Forest**
- **Support Vector Machine**
- **K-Nearest Neighbors**
- **Naïve Bayes**
- **K-Means Clustering**

Each algorithm was evaluated using suitable performance metrics

3. Deep Learning Models

Deep learning concepts were explored through practical implementation using TensorFlow and Keras.

Activities included:

- **Building Artificial Neural Networks**
- **Understanding CNN architecture**
- **Training deep learning models**
- **Evaluating prediction accuracy**
- **Optimizing model performance**

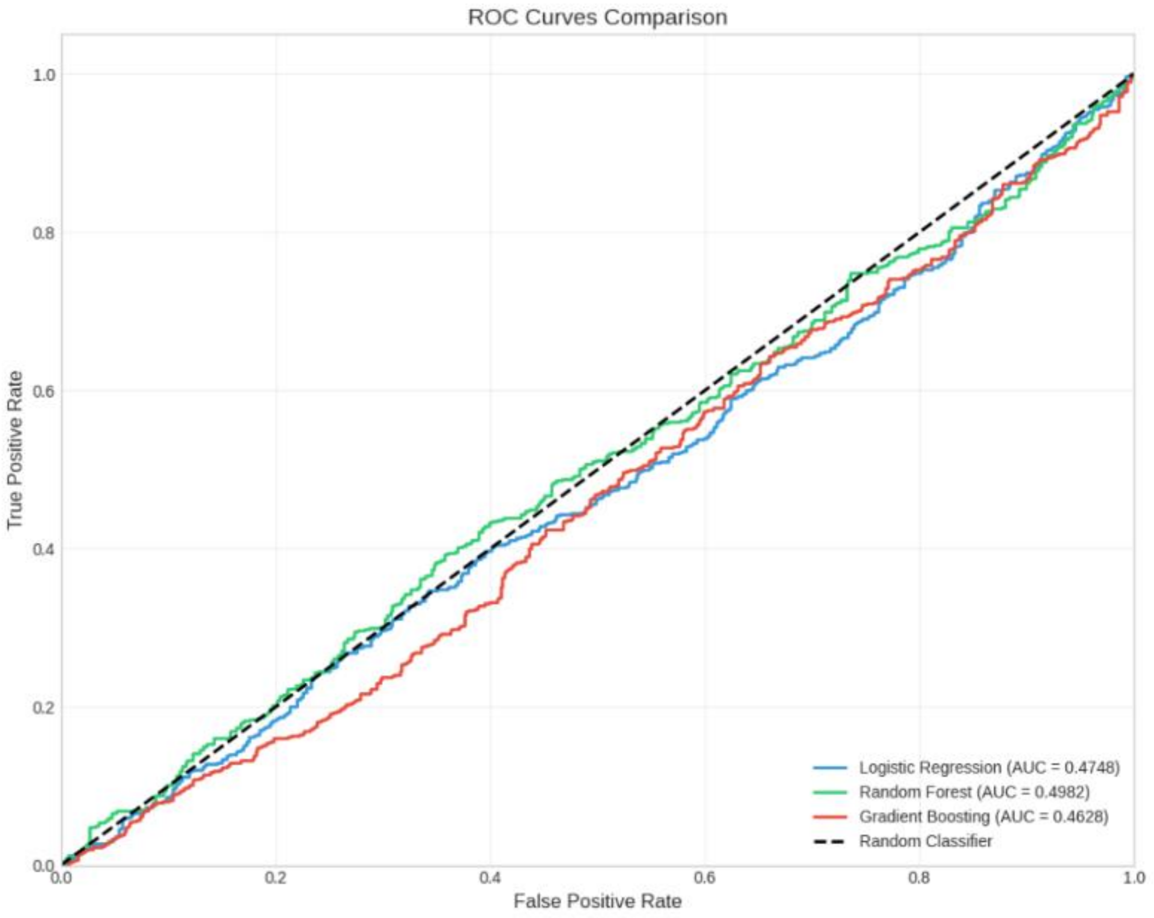
4. Data Analysis

Datasets were analyzed using Pandas and NumPy.

Tasks performed included:

- **Data cleaning**
- **Handling missing values**
- **Feature engineering**
- **Data transformation**
- **Statistical analysis**
- **Data visualization**

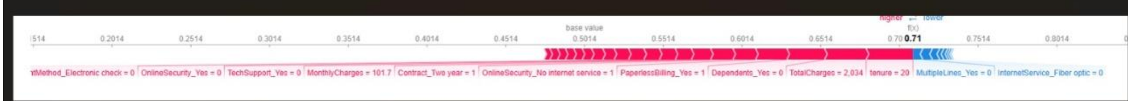
Proper preprocessing significantly improved model performance.

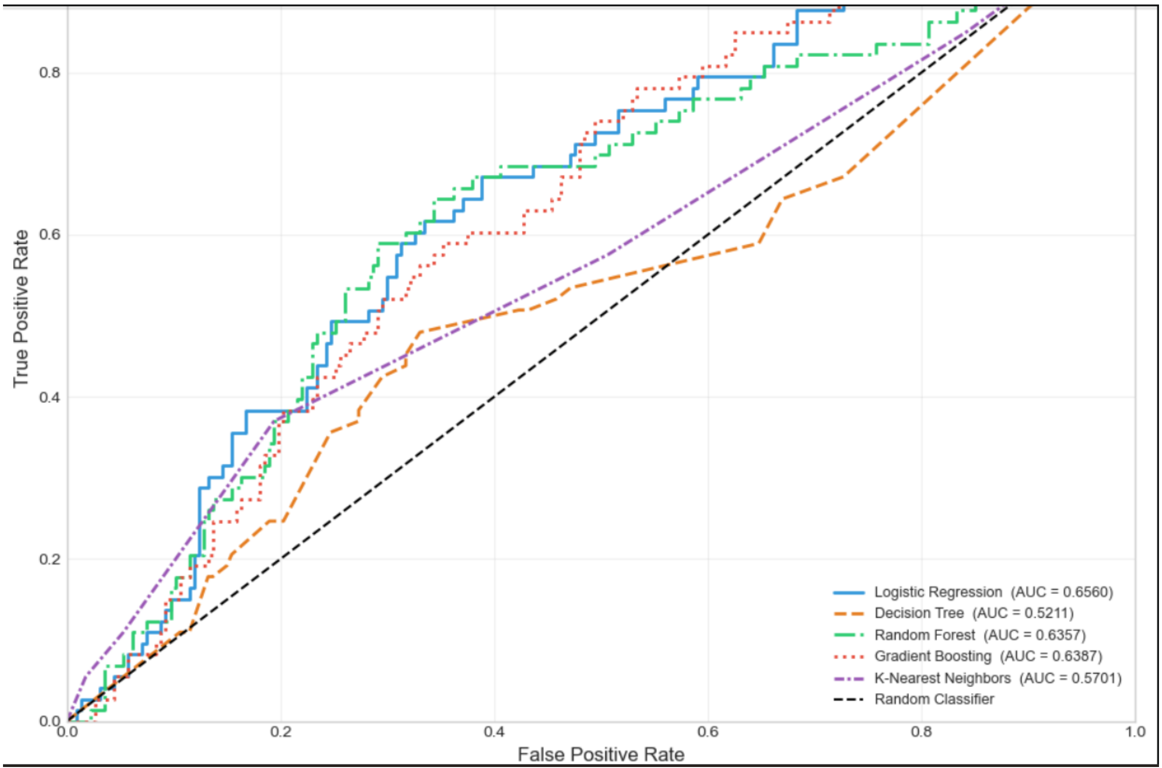


```python
        if prediction == 1:
            print(f"⚠ HIGH RISK: This customer is LIKELY TO CHURN.")
            print(f"Risk Probability: {probability:.2%}")
        else:
            print(f"✅ SAFE: This customer is LIKELY TO STAY (No Churn).")
            print(f"Churn Probability: {probability:.2%}")

        print("-"*50 + "\n")

    except Exception as e:
        print(f"\nAn error occurred during prediction. Please ensure the model pipeline is trained. Det

# Run the interactive function using our best trained pipeline from Step 6
test_custom_prediction(best_pipeline)
```

Python

```
==================================================
          TEST MODEL WITH CUSTOM DATA
==================================================
Enter customer details below (Press ENTER to keep the default value):
1. Tenure in months (Default: 12): 65
2. Monthly Charges in $ (Default: 75.0): 89
3. Contract [Month-to-month, One year, Two year] (Default: Month-to-month): Two year

--------------------------------------------------
PREDICTION RESULT:
--------------------------------------------------
✅ SAFE: This customer is LIKELY TO STAY (No Churn).
Churn Probability: 49.29%
--------------------------------------------------
```





CONCLUSION

The Artificial Intelligence internship provided an excellent opportunity to bridge the gap between academic learning and practical industry applications. Throughout the internship, comprehensive knowledge was gained in Artificial Intelligence, Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, and Data Science. The internship successfully combined theoretical concepts with hands-on implementation, enabling the development of intelligent systems capable of solving real-world problems.

One of the major achievements of the internship was learning the complete Artificial Intelligence development lifecycle. This included understanding problem statements, collecting datasets, preprocessing

data, selecting suitable machine learning algorithms, training predictive models, evaluating their performance, and interpreting the results. Working with real datasets enhanced analytical thinking and demonstrated the importance of data quality in building reliable AI solutions.

The internship also strengthened programming skills through the extensive use of Python and industry-standard libraries such as NumPy, Pandas, Matplotlib, Scikit-learn, TensorFlow, Keras, OpenCV, and NLTK. These tools enabled the implementation of supervised learning, unsupervised learning, and deep learning models while providing valuable exposure to data visualization, image processing, and natural language understanding.

Practical activities such as feature engineering, model optimization, and performance evaluation helped improve technical competency and problem-solving abilities. Different algorithms were compared using evaluation metrics including Accuracy, Precision, Recall, F1-Score, Mean Squared Error, and Confusion Matrix. This experience provided insight into selecting appropriate models based on specific problem requirements.

Beyond technical knowledge, the internship contributed significantly to professional development. Skills such as teamwork, communication, technical documentation, project planning, presentation, and time management were strengthened through continuous interaction with mentors and participation in practical assignments.

The internship also highlighted the importance of ethical Artificial Intelligence by emphasizing fairness, transparency, accountability, privacy, and responsible AI development. Understanding these principles is essential for building AI systems that benefit society while minimizing risks.

Overall, the internship successfully achieved its objectives and provided a strong foundation for future learning and professional growth in Artificial Intelligence. The knowledge and practical experience gained during this internship will be valuable for higher studies, research, and careers in Artificial Intelligence, Machine Learning, Data Science, Software Engineering, and Intelligent Automation.

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The following references were consulted to understand Artificial Intelligence concepts and practical implementation.

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